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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	overinfinity
Selected Domain	Domain 1 - The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	4.4 (primary), 4.3, 8.5, 8.2
Project / Solution Name	AURA-Workforce
Name (Leader)	Ahmad Adam Hakimi bin Naeman
Email & Phone (Leader)	0damkim254@gmail.com / +60 18-344 5135
University Affiliation(s)	International Islamic University Malaysia (IIUM)
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's workforce is being reshaped by AI faster than its training system can respond. 56% of Malaysian professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025). The Ministry of Science, Technology and Innovation estimates that more than 600,000 workers will need reskilling within three to five years.

The people affected are specific: a procurement assistant whose routine processing is being automated, a fresh graduate choosing a first role, an HR manager planning next year's training budget, a TVET career officer advising students from a printed list of jobs.

What they lack is not information but trustworthy direction. Career advice today is generic, static and rarely sourced. A worker cannot see which roles are rising or declining, does not know which existing skills transfer, cannot judge how long a pathway takes against the hours they actually have, and has no way to check where a claim came from. Counsellors work from frameworks that are out of date by the time they are published.

The cost is paid twice: individuals buy courses that do not lead to work, and employers retrain people for roles that are disappearing. Left unaddressed, the AI transition widens the gap between those who can afford continuous reskilling and those who cannot.

This is a skills-transition problem before it is a technology problem, which is why it maps directly to SDG 4.4, 4.3, 8.5 and 8.2.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AURA-Workforce is a privacy-first career-transition planner. A worker enters their current role, skills, experience, available weekly hours and target sectors. The system returns ranked future-ready roles with an explainable match score, a skill-gap breakdown, a sequenced reskilling pathway with hours and cost notes, and an explicit mapping of every role and module to an SDG target.

AI is applied at three points, each chosen because a model or a live feed beats a static rule:

1. Skill normalisation. Form and free-text input are mapped onto a canonical skill vocabulary, so 'chasing POs in Excel' is recognised as procurement plus spreadsheet analysis instead of being missed as unknown text.
2. Live trend layer. Public job postings are fetched from an open API, counted against each role's required skills, and converted into a dated demand index. Every figure carries its source, the window it covers and the sample size, and the interface states when the data was last refreshed.
3. Scenario modelling. Two disruption scenarios (moderate: four-year runway; aggressive: eighteen-month runway) change planning urgency instead of pretending to predict the future.

Deliberately, the ranking itself is deterministic and inspectable: weights, matched skills and gaps are visible, so a counsellor can challenge any recommendation. The model never invents a role, a course, a provider or a statistic; every course link resolves to a real provider page.

Technical skills applied: language-model API calls for text understanding, React and Vite deployed on Replit, and a Postgres/Drizzle data layer for saved plans in Phase 2.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, Vite 7, Tailwind CSS 4, TanStack Query, Wouter routing, TypeScript 5.9.
Backend: Node.js 24, Express 5, Zod validation for every response.
Contract: OpenAPI 3.1 specification with Orval code generation for typed client hooks and Zod schemas.
Data: curated public snapshot (U.S. BLS Occupational Outlook Handbook, WEF Future of Jobs 2025) plus a live public job-posting feed (Arbeitnow Job Board API, no key required). Postgres + Drizzle reserved for saved plans in Phase 2.
Hosting: Replit (build and deployment), single public URL, secrets held in Replit Secrets.
System diagram: browser (React SPA) -> /api (Express) -> [curated catalogue | live trend layer] -> deterministic scoring engine -> ranked roles, gaps and pathway. The trend layer is the only component that changes without a redeploy.
Sponsor resources: Replit Core (one month per member) is used for hosting and deployment of the prototype. The RunPod credit (USD 20 per team) is reserved for Phase 2, where the same model endpoint will be self-hosted on GPU. The live trend feed is a free public API. All of the above is declared in the Technical & Tools Declaration.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We started from people, not technology: the four personas in the domain brief (displaced worker, career switcher, HR/L&D manager, TVET counsellor) and one moment of need, the week a worker learns their role is changing. Conversations with peers and working adults showed the same pattern: people do not want a prediction, they want to know what to do next and whether to trust it.
That produced three constraints that shaped every screen: show the source, show the uncertainty, and let the user change the assumption. The moderate/aggressive scenario control exists because a single forecast would be dishonest.
We prototyped in the browser daily and deployed to Replit from the first day, so feedback was given on the real artifact rather than a mockup. All testing used synthetic sample profiles, in line with the rules on

personal data. Weaknesses found in testing (opaque scores, unexplained gaps, small text in dense panels) were fixed in the same week.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	4.4.1 (youth and adults with relevant skills for employment), 4.3.1 (access to affordable lifelong learning), 8.5.2 (unemployment rate), 8.2.1 (productivity growth). Primary target: 4.4.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	56% of Malaysian professionals use AI at work but only 26% received formal training (Hays Asia Salary Guide 2025); MOSTI projects more than 600,000 workers needing reskilling within 3-5 years. In today's career-advice market, recommendations are unsourced and static: our own baseline reflects this, with 6 curated market signals but only 1 updatable trend source, and no dated evidence attached to individual role or course recommendations.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1) Percentage of role, skill and course recommendations carrying a verifiable source and date. 2) Median time from profile start to a ranked pathway. 3) Number of independent, updatable trend sources feeding role ranking.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	100% of recommendations sourced and dated (up from approximately 60% at prototype stage); median time-to-first-pathway under 5 minutes (currently about 20 minutes when done manually with a counsellor); at least 2 independent trend sources, including one Malaysian source, so the ranking visibly responds to regional demand.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19 / React DOM	code	MIT
Vite 7	code	MIT
Tailwind CSS 4	code	MIT
Radix UI primitives	code	MIT
TanStack Query 5	code	MIT
Express 5	code	MIT
Zod 3	code	MIT
Orval (codegen)	code	MIT
Lucide icons	code	ISC
Arbeitnow Job Board API	dataset	Public API, free, no key; source cited in-app

U.S. BLS Occupational Outlook Handbook	dataset	Public domain (U.S. Government work)
WEF Future of Jobs Report 2025	dataset	Used for quoted figures with attribution

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Ahmad Adam Hakimi bin Naeman (Team Leader, overinfinity)

23 October 2026